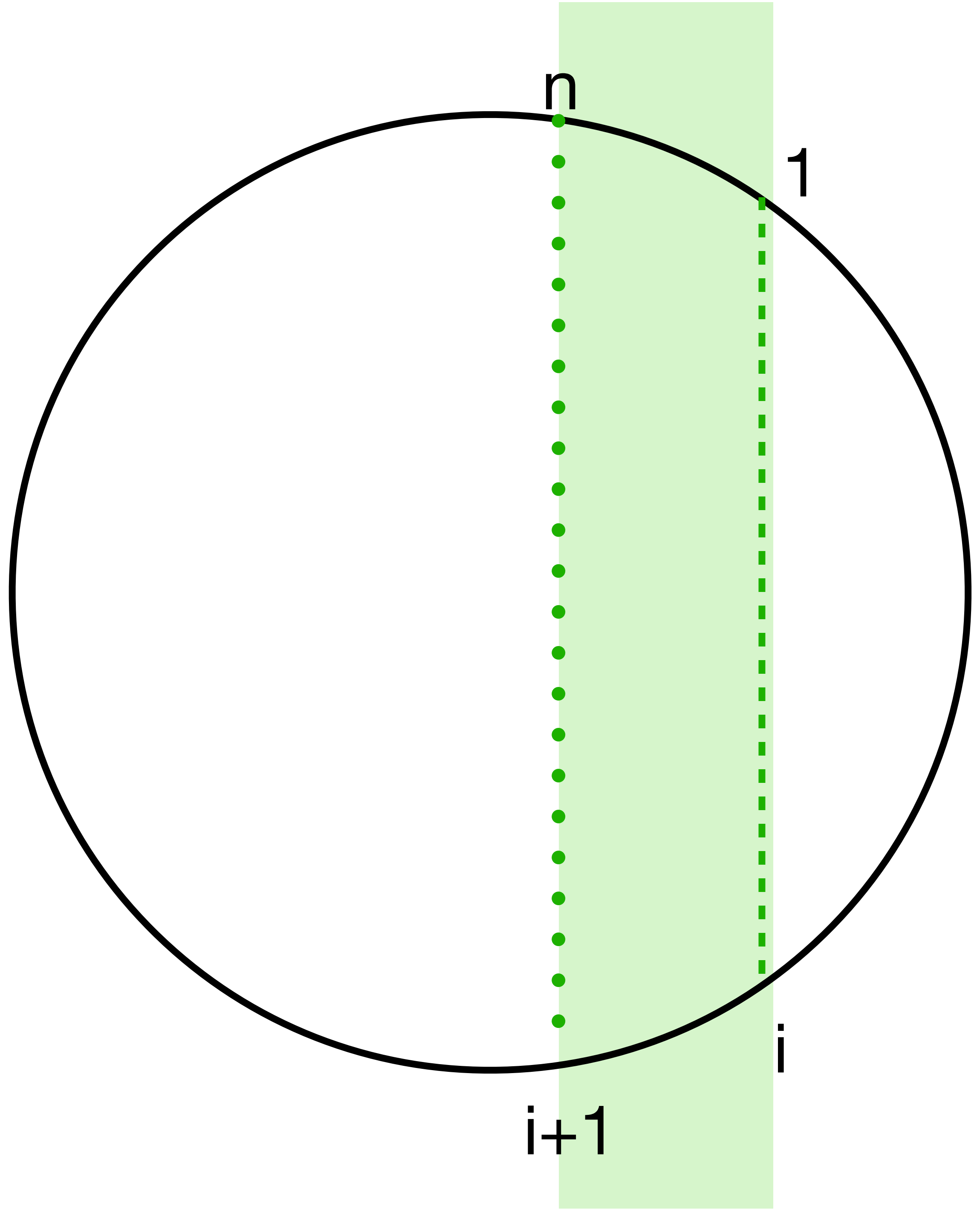
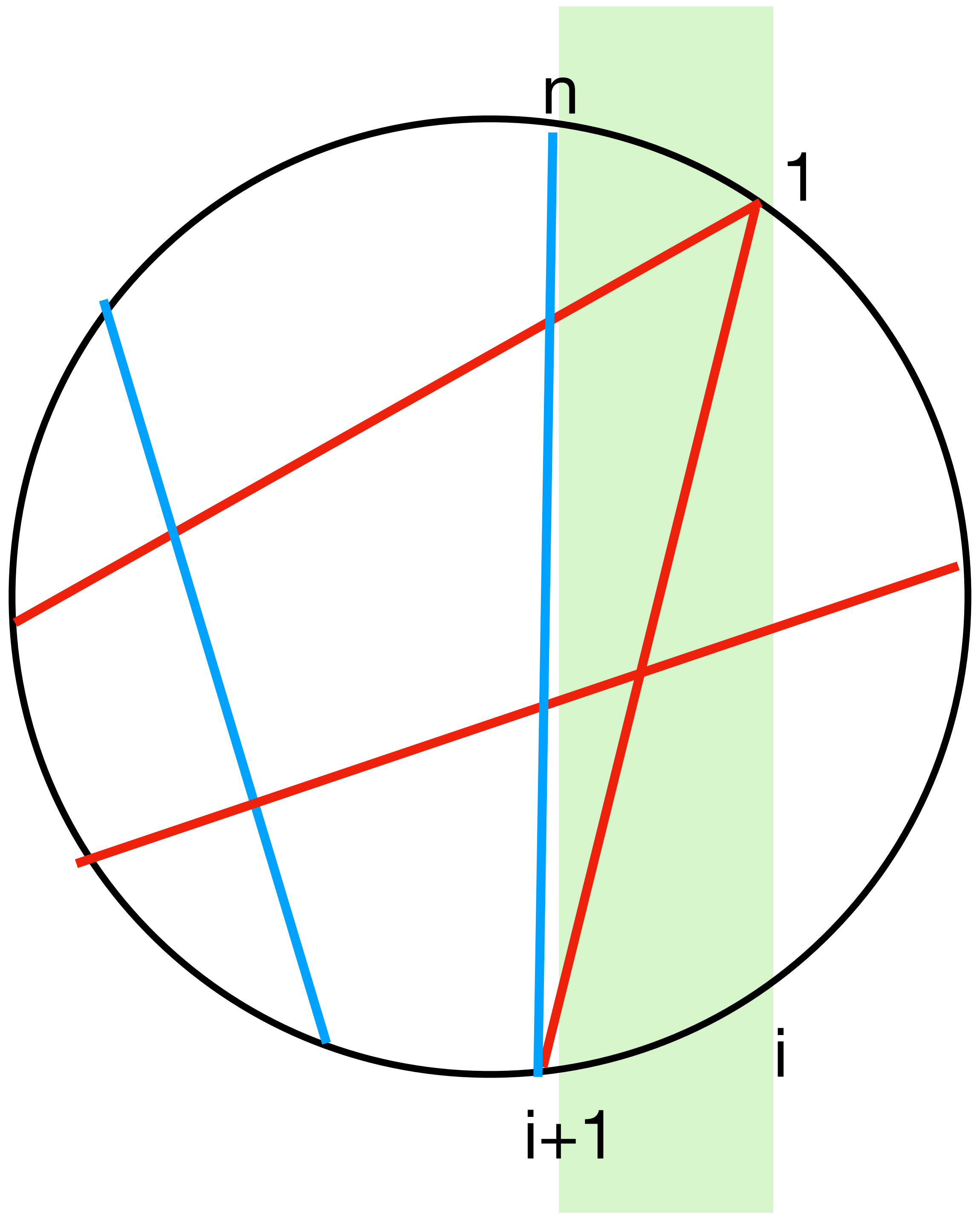
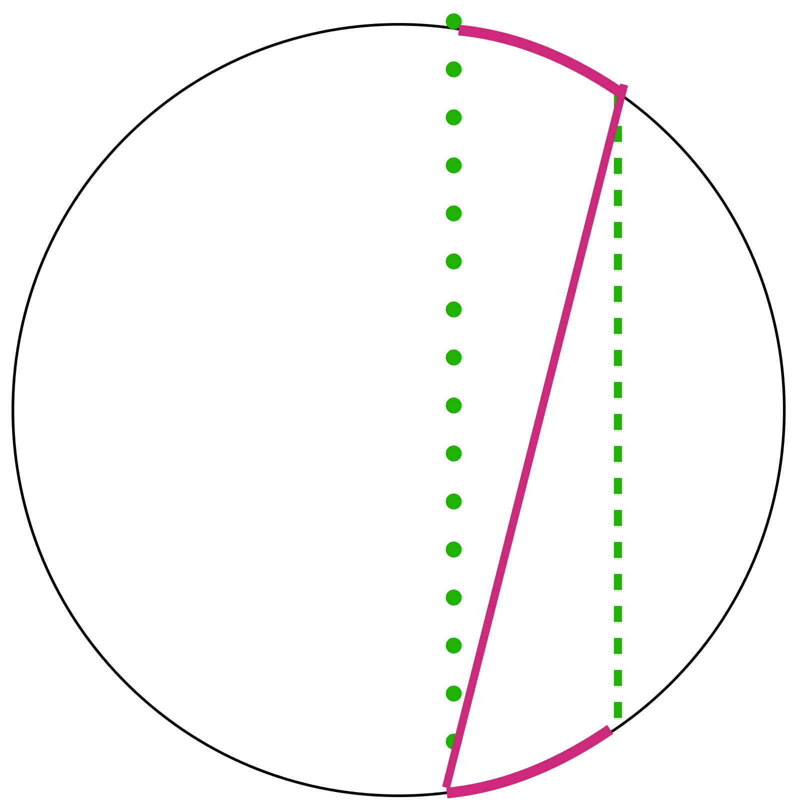
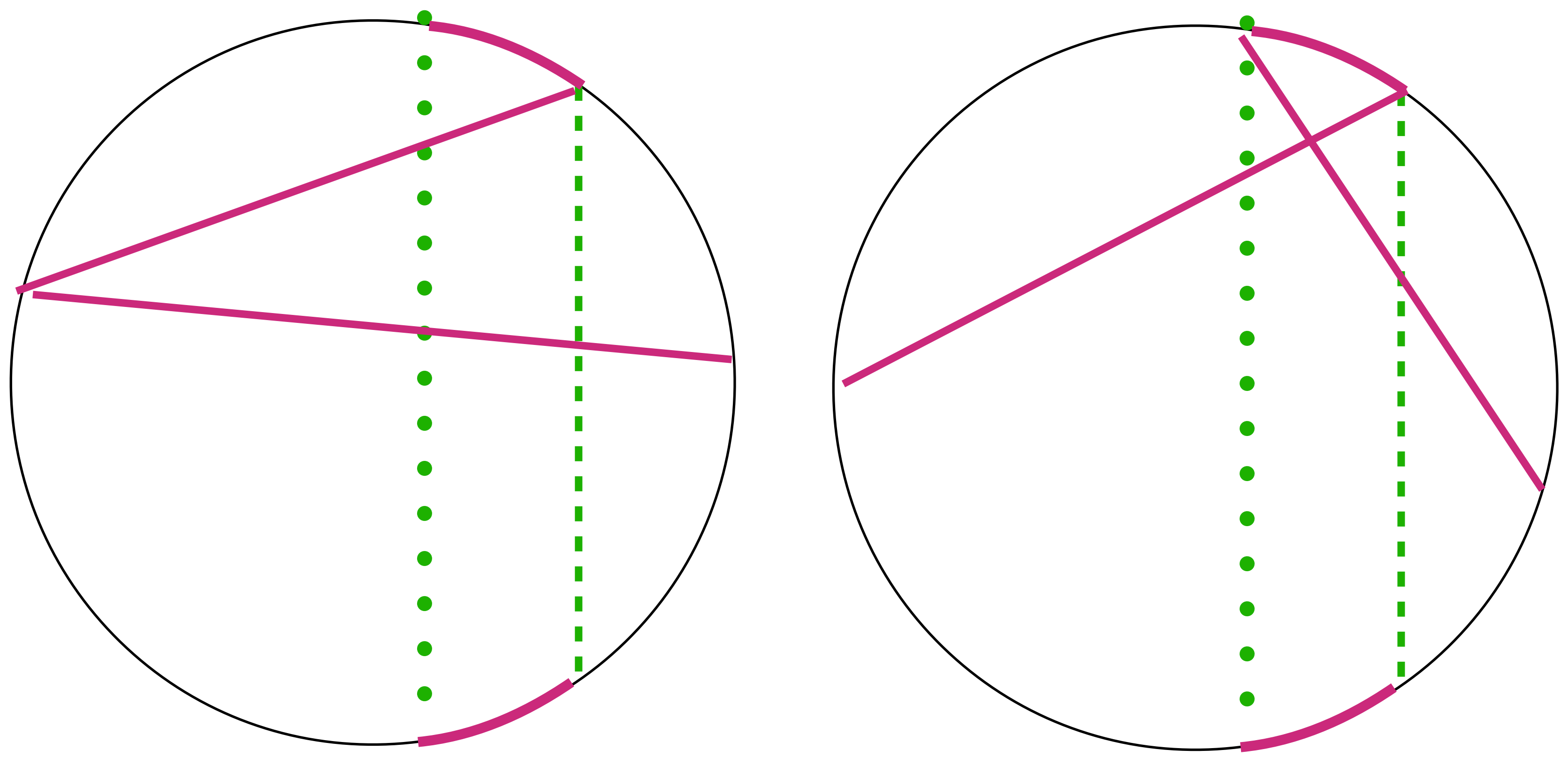




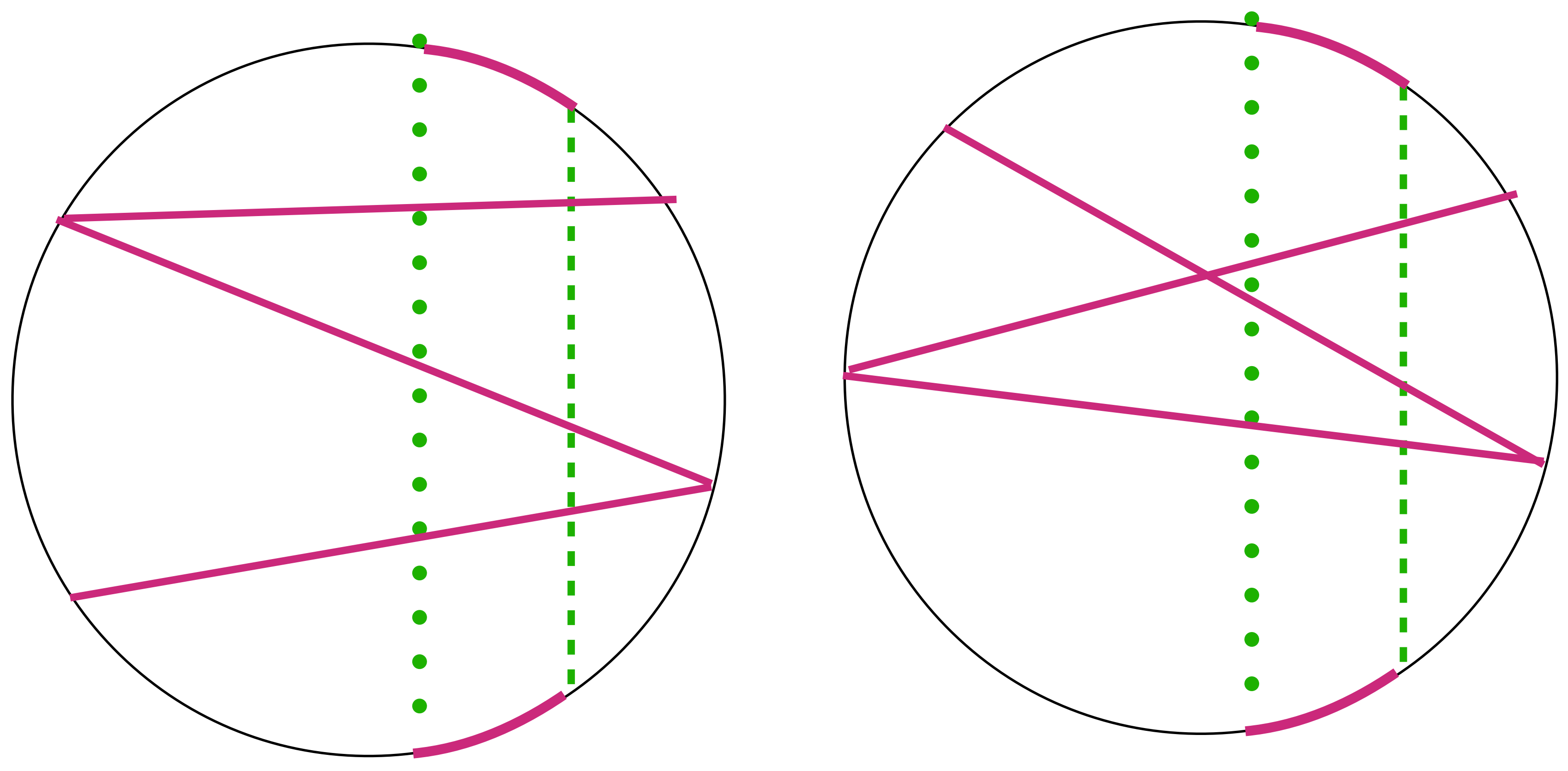
- Consider the zero associated to chord $X_{1i} \leftrightarrow Z_{1i} : c_{a,b} = 0,\ a \in 1,\ldots, i-1$
- the zero Z_{1i} defines the enclosing chord X_{1i} and the complement $X_{i+1,n}$
- a chord appears in the Z_{1i} conditions iff it crosses the band between X_{1i} and $X_{i+1,n}$
  red=crosses; blue=does not cross
- the logic: if there is no zig-zag across the band, Z_{1i} is a killing zero (important: the edges X_{1n} and $X_{i,i+1}$ are both automatically present)
  if a zig-zag is present, it always forces the closing chord to vanish



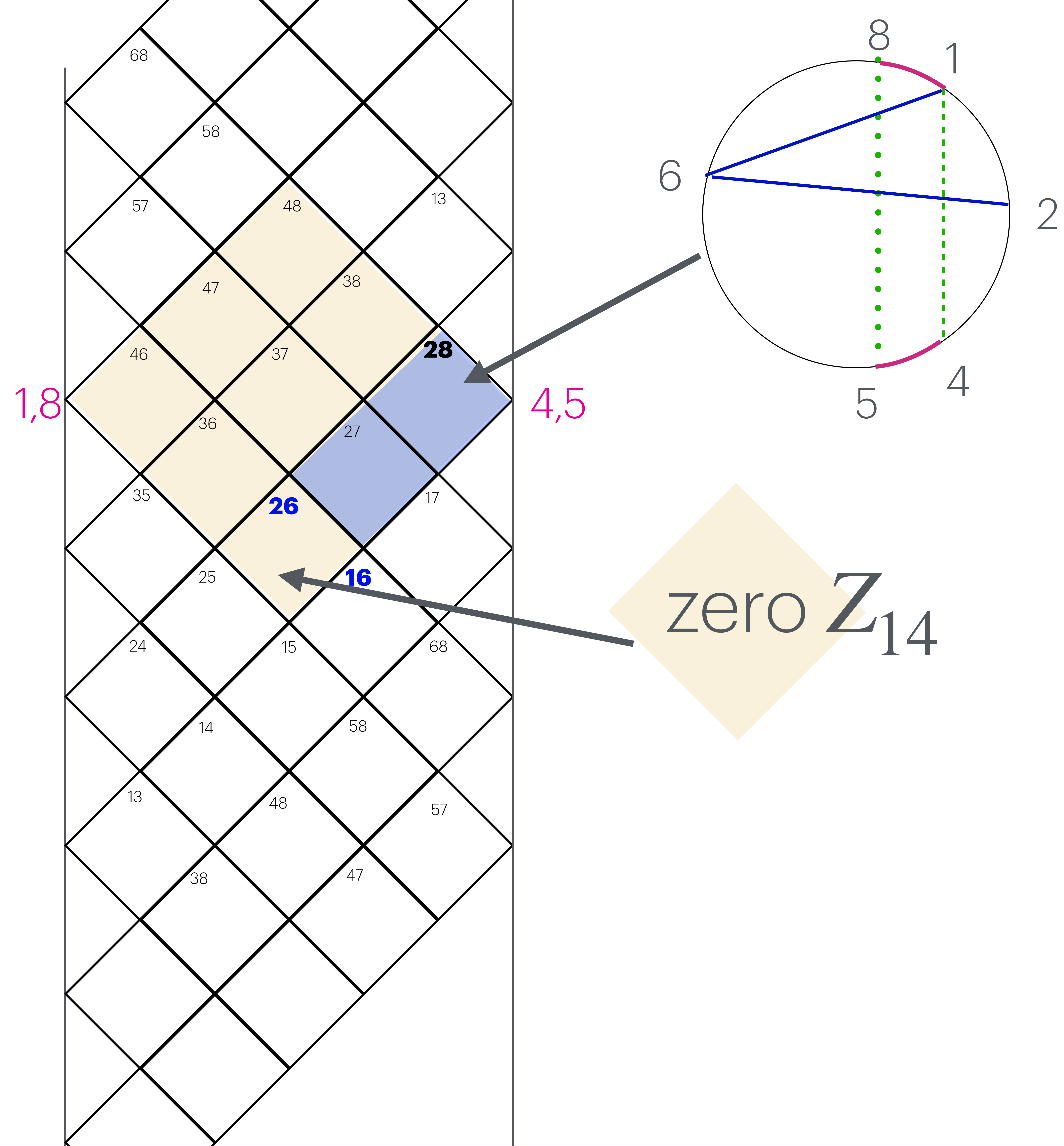
simplest “1 line” zig-zag
-uses both edges
-this gives a simple condition: the zero chord must be a boundary of a non-triangulated region



“2-line” zig-zag
(uses 1 edge)



“3 line” zig-zag
(uses no edge)



the three lines forming a zig-zag force $X_{28} = 0$

on the mesh, this means the three lines form the blue rectangle.
the zero conditions force the 4th corner to vanish